THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

ISE 430/530: Manufacturing Systems and Facilities Design

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MIP-Parameters

B_x=building length
B_y=building width
A_i=area of department *i*
L_i^l=lower limit on the length of department *i*
L_i^u=upper limit on the length of department *i*
W_i^l=lower limit on the width of department *i*
W_i^u=upper limit on the width of department *I*

Decision Variables

α_i = *x*-coordinate of the centroid of department *i*
β_i = y-coordinate of the centroid of department *i*
x'_i = *x*-coordinate of the left (or west) side of department *i*
x''_i = *x*-coordinate of the right (or east) side of department *i*
y'_i = *y*-coordinate of the bottom (or south) side of department *i*
y''_i = *y*-coordinate of the top (or north) side of department *i*
z^x_{ij}= equal to 1 if department *i* is strictly to the east of department *j*
z^y_{ij}= equal to 1 if department *i* is strictly to the north of department *j*

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MIP-Model

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$ **nonlinear**

subject to :

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \quad \text{for all } i$$ **nonlinear**

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

$$\text{If } z_{ij}^x = 1 \Rightarrow x_j'' \le x_i' \quad \text{for all } i \text{ and } j, i \ne j$$ **conditional**

$$\text{If } z_{ij}^y = 1 \Rightarrow y_j'' \le y_i' \quad \text{for all } i \text{ and } j, i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j, i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y_i', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y \; 0/1 \quad \text{for all } i \text{ and } j, i \ne j$$

“Conditional” Constraints

Consider the two constraints:

$$\text{if } f(x_1, x_2, x_3 \ldots, x_n) = 1 \Rightarrow g(x_1, x_2, x_3 \ldots, x_n) \leq 0$$

$$f(x_1, x_2, x_3 \ldots, x_n) = \{0,1\}$$

Suppose we wish to ensure that $g(x_1, x_2, \mathrm{K}\ , x_n) \leq 0$ **only when**

$$f(x_1, x_2, x_3 \ldots, x_n) = 1$$

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“Conditional” Constraints

Then, adding the following constraints to the formulation will ensure this:

$$g(x_1, x_2, x_3 \ldots, x_n) \leq (1 - f(x_1, x_2, x_3 \ldots, x_n))M$$

where M = very large number

CASE1

$$f(x_1, x_2, x_3 \ldots, x_n) = 1;$$

$$g(x_1, x_2, x_3 \ldots, x_n) \leq 0$$

Condition holds

CASE2

$$f(x_1, x_2, x_3 \ldots, x_n) = 0;$$

$$g(x_1, x_2, x_3 \ldots, x_n) \leq M$$

Condition does not holds

MIP-Model

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$ **nonlinear**

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \quad \text{for all } i$$ **nonlinear**

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

$$x_j'' \le x_i' + (1 - z_{ij}^x)M \quad \text{for all } i \text{ and } j,\ i \ne j$$ **conditional**

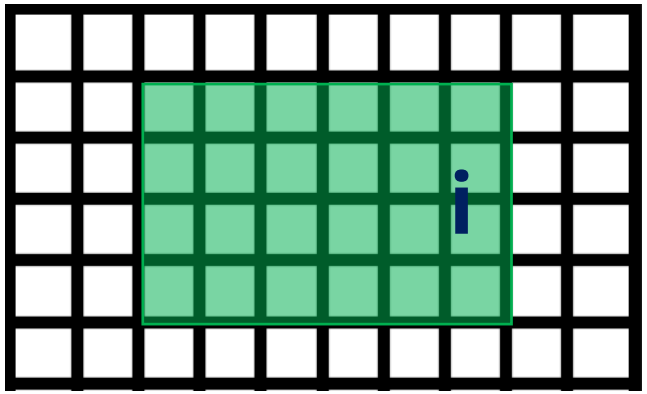
$$y_j'' \le y_i' + (1 - z_{ij}^y)M \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j,\ i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y\, 0/1 \quad \text{for all } i \text{ and } j,\ i \ne j$$



Approximation of area

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$

$$(x_i'' - x_i')(y_i'' - y_i') = 24$$

$$10 \le (x_i'' - x_i' + y_i'' - y_i') \le 11$$

Converting "Absolute" value as a linear expression

Consider the following objective function:

$$\text{minimize} \sum_{j \in J} c_j |x_j|$$

Replace $x_j = x_j^+ - x_j^-$ where $|x_j| = x_j^+ + x_j^-$ and $x_j^+, x_j^- \geq 0$

CASE1 $x_j \geq 0$

$$x_j = x_j^+ - x_j^-$$

$$x_j^+ = x_j, x_j^- = 0$$

Non-negativity satisfied

CASE2 $x_j \leq 0$

$$x_j = x_j^+ - x_j^-$$

$$x_j^+ = 0, x_j^- = -x_j$$

Non-negativity satisfied

Converting "Absolute" value as a linear expression

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(|\alpha_i - \alpha_j| + |\beta_i - \beta_j| \right)$$ **nonlinear**

subject to :

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$ **nonlinear**

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

$$x_j'' \le x_i' + (1 - z_{ij}^x)M \quad \text{for all } i \text{ and } j, i \ne j$$

$$y_j'' \le y_i' + (1 - z_{ij}^y)M \quad \text{for all } i \text{ and } j, i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j, i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y 0/1 \quad \text{for all } i \text{ and } j, i \ne j$$

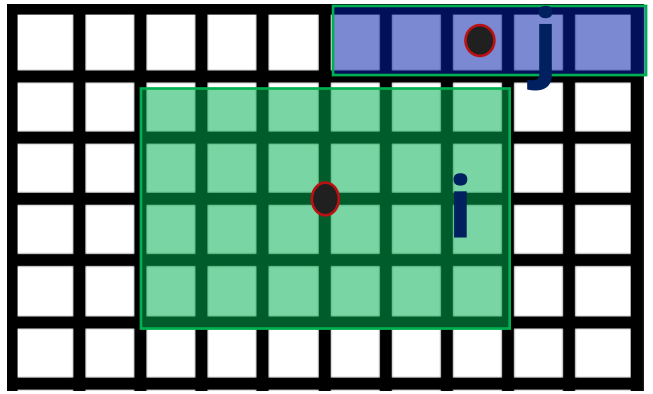
$$\alpha_i - \alpha_j = \alpha_{ij}^+ - \alpha_{ij}^-$$

$$\beta_i - \beta_j = \beta_{ij}^+ - \beta_{ij}^-$$

$$|\alpha_i - \alpha_j| = \alpha_{ij}^+ + \alpha_{ij}^- \qquad |\beta_i - \beta_j| = \beta_{ij}^+ + \beta_{ij}^-$$

$$\alpha_{ij}^+, \alpha_{ij}^-, \beta_{ij}^+, \beta_{ij}^- \ge 0$$

$\alpha_i = 5, \beta_i = 3$ $\alpha_j = 7.5, \beta_j = 5.5$



$$|\alpha_i - \alpha_j| = |5 - 7.5| = 2.5$$

$$-2.5 = \alpha_i - \alpha_j = \alpha_{ij}{}^+ - \alpha_{ij}{}^-$$

$$2.5 = |\alpha_i - \alpha_j| = \alpha_{ij}{}^+ + \alpha_{ij}{}^-$$

$$\alpha_{ij}{}^+ = 0, \alpha_{ij}{}^- = 2.5$$

Converting "Absolute" value as a linear expression

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\alpha_{ij}^+ + \alpha_{ij}^- + \beta_{ij}^+ + \beta_{ij}^- \right)$$

nonlinear

subject to :

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

$$x_j'' \le x_i' + (1 - z_{ij}^x)M \quad \text{for all } i \text{ and } j, i \ne j$$

$$y_j'' \le y_i' + (1 - z_{ij}^y)M \quad \text{for all } i \text{ and } j, i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j, i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y \, 0/1 \quad \text{for all } i \text{ and } j, i \ne j$$

Additional constraints

$$\alpha_i - \alpha_j = \alpha_{ij}^+ - \alpha_{ij}^-$$

$$\beta_i - \beta_j = \beta_{ij}^+ - \beta_{ij}^-$$

$$\alpha_{ij}^+, \alpha_{ij}^-, \beta_{ij}^+, \beta_{ij}^- \ge 0$$

Converting "Absolute" value as a linear expression

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\alpha_{ij}^+ + \alpha_{ij}^- + \beta_{ij}^+ + \beta_{ij}^- \right)$$

subject to :

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

$$x_j'' \le x_i' + (1 - z_{ij}^x)M \quad \text{for all } i \text{ and } j, i \ne j$$

$$y_j'' \le y_i' + (1 - z_{ij}^y)M \quad \text{for all } i \text{ and } j, i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j, i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y 0/1 \quad \text{for all } i \text{ and } j, i \ne j$$

$$\alpha_i - \alpha_j = \alpha_{ij}^+ - \alpha_{ij}^-$$

$$\beta_i - \beta_j = \beta_{ij}^+ - \beta_{ij}^-$$

$$\alpha_{ij}^+, \alpha_{ij}^-, \beta_{ij}^+, \beta_{ij}^- \ge 0$$

MIP-Model: Example

Assume 3 departments with space requirements: A (3000) B (1000) C (4000) with building of 100 by 80.

	A	B	C
A	-	10	20
B	-	-	50
C	-	-	-

Flow between chart

α_i = ***x*-coordinat of the centroid of department *i*** $\alpha_1\ \alpha_2\ \alpha_3$

β_i = **y-coordinate of the centroid of department *I*** $\beta_1\ \beta_2\ \beta_3$

x'_i = ***x*-coordinate of the left (or west) side of department *I*** $x'_1\ x'_2\ x'_3$

x''_i = ***x*-coordinate of the right (or east) side of department *i*** $x''_1\ x''_2\ x''_3$

y'_i = ***y*-coordinate of the bottom (or south) side of department *i*** $y'_1\ y'_2\ y'_3$

y''_i = ***y*-coordinate of the top (or north) side of department *i*** $y''_1\ y''_2\ y''_3$

z^x_{ij}= **equal to 1 if department *i* is strictly to the east of department *j***

$z^x_{12}, z^x_{13}, z^x_{21}, z^x_{23}, z^x_{31}, z^x_{32}$: ***binary***

z^y_{ij}= **equal to 1 if department *i* is strictly to the north of department *j***

$z^y_{12}, z^y_{13}, z^y_{21}, z^y_{23}, z^y_{31}, z^y_{32}$: ***binary***

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MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

	A	B	C
A	-	10	20
B	-	-	50
C	-	-	-

B_x=building length B_x=100

B_y=building width B_y=80

A_i=area of department *i*

A_1=3000, A_2=1000, A_3=4000

L_i^l=lower limit on the length of department *I*

Assume: L_1^l=37.5, L_2^l=12.5, L_3^l=50

L_i^u=upper limit on the length of department *I*

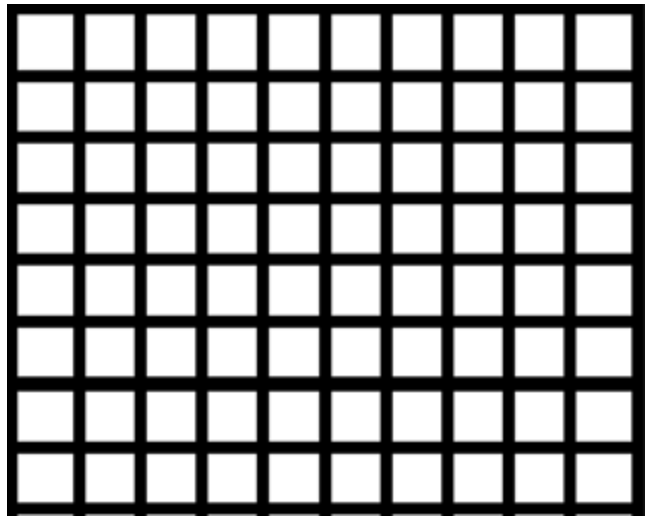
Assume: L_1^u=100, L_2^u=100, L_3^u=100

W_i^l=lower limit on the width of department *I*

Assume: W_1^l=30, W_2^l=10, W_3^l=40

W_i^u=upper limit on the width of department *I*

Assume: W_1^u=80, W_2^u=80, W_3^u=80



MIP-Model: Example

	A	B	C
A	-	10	20
B	-	-	50
C	-	-	-

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(|\alpha_i - \alpha_j| + |\beta_i - \beta_j| \right)$$

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

Minimize $\sum_i \sum_j f_{ij} (|\alpha_i - \alpha_j| + |\beta_i - \beta_j|)$

$= 10(|\alpha_1 - \alpha_2| + |\beta_1 - \beta_2|)$

$+20(|\alpha_1 - \alpha_3| + |\beta_1 - \beta_3|)$

$+50(|\alpha_2 - \alpha_3| + |\beta_2 - \beta_3|)$

$$10(\alpha_{12}^+ + \alpha_{12}^- + \beta_{12}^+ + \beta_{12}^-)$$

$$+20(\alpha_{12}^+ + \alpha_{13}^- + \beta_{13}^+ + \beta_{13}^-)$$

$$+50(\alpha_{22}^+ + \alpha_{32}^- + \beta_{23}^+ + \beta_{23}^-)$$

Length restriction for department

L_1^l=37.5, L_2^l=12.5, L_3^l=50

L_1^u=100, L_2^u=100, L_3^u=100

$37.5 \le x_1'' - x_1', x_1'' - x_1' \le 100$

$12.5 \le x_2'' - x_2', x_2'' - x_2' \le 100$

$50 \le x_3'' - x_3', x_3'' - x_3' \le 100$

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \quad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \quad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \quad \text{for all } i$$

	A	B	C
A	-	10	20
B	-	-	50
C	-	-	-

Width restriction for department

W_1^l=30, W_2^l=10, W_3^l=40

W_1^u=80, W_2^u=80, W_3^u=80

$$30 \le y_1'' - y_1', y_1'' - y_1' \le 80$$

$$10 \le y_2'' - y_2', y_2'' - y_2' \le 80$$

$$40 \le y_3'' - y_3', y_3'' - y_3' \le 80$$

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

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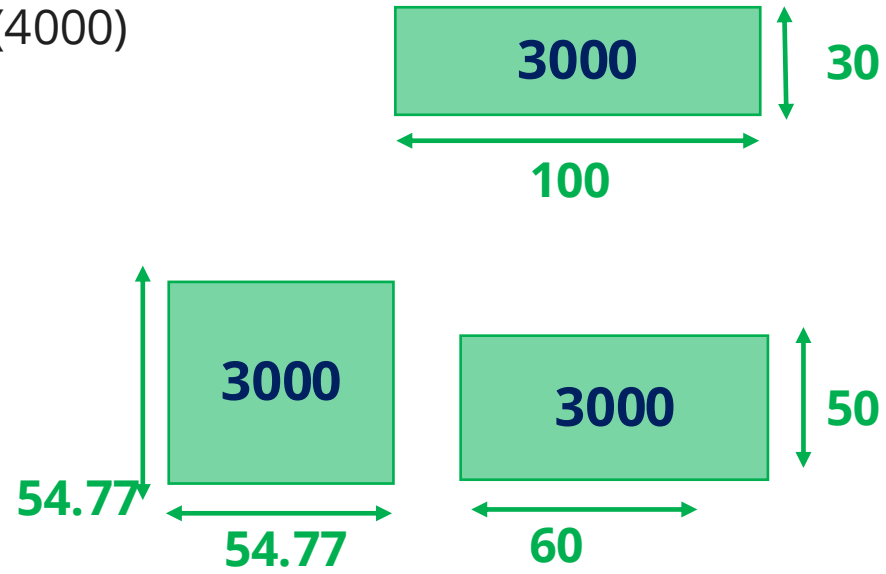
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$$\alpha_i = 0.5 x_i' + 0.5 x'' \qquad \text{for all } i$$

$$\beta_i = 0.5 y_i' + 0.5 y_i'' \qquad \text{for all } i$$



A_1=3000, A_2=1000, A_3=4000

Space Requirement

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \qquad \text{for all } i$$

$$109 \le (x_1'' - x_1' + y_1'' - y_1') \le 130$$

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

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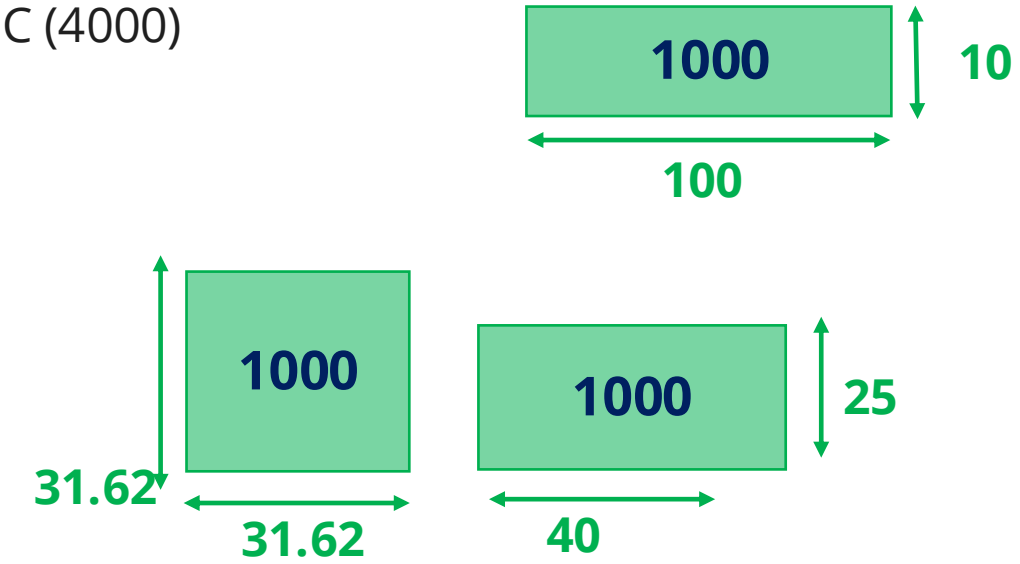
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$$\alpha_i = 0.5x_i' + 0.5x'' \qquad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \qquad \text{for all } i$$



A_1=3000, A_2=1000, A_3=4000

Space Requirement

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \qquad \text{for all } i$$

$$63 \le (x_2'' - x_2' + y_2'' - y_2') \le 110$$

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MIP-Model: Example

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$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \quad \text{for all } i$$

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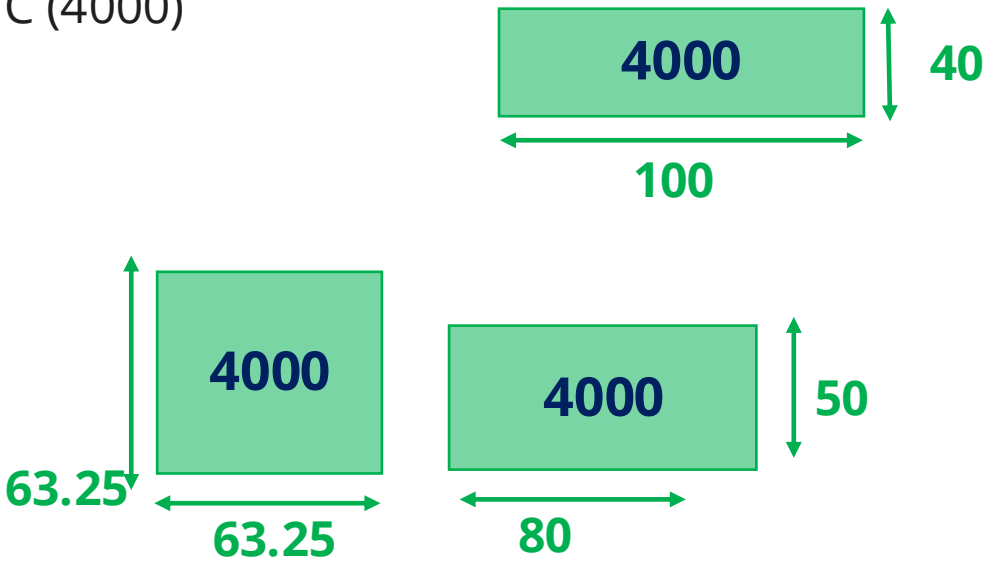
$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \quad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \quad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \quad \text{for all } i$$

$$\alpha_i = 0.5 x_i' + 0.5 x'' \quad \text{for all } i$$

$$\beta_i = 0.5 y_i' + 0.5 y_i'' \quad \text{for all } i$$



A_1=3000, A_2=1000, A_3=4000

Space Requirement

$$(x_i'' - x_i')(y_i'' - y_i') = A_i \quad \text{for all } i$$

$$126 \le (x_3'' - x_3' + y_3'' - y_3') \le 140$$

THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000)
with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \qquad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \qquad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \qquad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \qquad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \qquad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \qquad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \qquad \text{for all } i$$

$$109 \le (x_1'' - x_1' + y_1'' - y_1') \le 130$$

$$63 \le (x_2'' - x_2' + y_2'' - y_2') \le 110$$

$$126 \le (x_3'' - x_3' + y_3'' - y_3') \le 140$$

Space Requirement

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \qquad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \qquad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \qquad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \qquad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \qquad \text{for all } i$$

Range

$$\alpha_i = 0.5x_i' + 0.5x'' \qquad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \qquad \text{for all } i$$

B_x=100

B_y=80

$$x_1' \le x_1'' \qquad y_1' \le y_1''$$

$$x_2' \le x_2'' \qquad y_2' \le y_2''$$

$$x_3' \le x_3'' \qquad y_3' \le y_3''$$

$$x_1'' \le 100 \qquad y_1'' \le 80$$

$$x_2'' \le 100 \qquad y_1'' \le 80$$

$$x_3'' \le 100 \qquad y_1'' \le 80$$

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000)
with building of 100 by 80.

$$z = \sum_i \sum_j f_{ij} c_{ij} \left(\left| \alpha_i - \alpha_j \right| + \left| \beta_i - \beta_j \right| \right)$$

subject to:

$$L_i^l \le (x_i'' - x_i') \le L_i^u \qquad \text{for all } i$$

$$W_i^l \le (y_i'' - y_i') \le W_i^u \qquad \text{for all } i$$

$$P_i^l \le (x_i'' - x_i' + y_i'' - y_i') \le P_i^u \qquad \text{for all } i$$

$$0 \le x_i' \le x_i'' \le B_x \qquad \text{for all } i$$

$$0 \le y_i' \le y_i'' \le B_y \qquad \text{for all } i$$

$$\alpha_i = 0.5x_i' + 0.5x'' \qquad \text{for all } i$$

$$\beta_i = 0.5y_i' + 0.5y_i'' \qquad \text{for all } i$$

$$\alpha_1 = 0.5x_1' + 0.5x_1''$$

$$\alpha_2 = 0.5x_2' + 0.5x_2''$$

$$\alpha_3 = 0.5x_3' + 0.5x_3''$$

$$\beta_1 = 0.5y_1' + 0.5y_1''$$

$$\beta_2 = 0.5y_2' + 0.5y_2''$$

$$\beta_3 = 0.5y_3' + 0.5y_3''$$

Centroid coordinates

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000)
with building of 100 by 80.

$$x''_j \le x'_i + (1 - z^x_{ij})M \qquad \text{for all } i \text{ and } j,\ i \ne j$$
$$y''_j \le y'_i + (1 - z^y_{ij})M \qquad \text{for all } i \text{ and } j,\ i \ne j$$
$$z^x_{ij} + z^x_{ji} + z^y_{ij} + z^y_{ji} \ge 1 \qquad \text{for all } i \text{ and } j,\ i < j$$
$$\alpha_i, \beta_i \ge 0 \qquad \text{for all } i$$
$$x'_i, x''_i, y', y''_i \ge 0 \qquad \text{for all } i$$
$$z^x_{ij}, z^y_{ij} = \{0,1\} \qquad \text{for all } i \text{ and } j,\ i \ne j$$
$$\alpha_i - \alpha_j = \alpha^+_{ij} + \alpha^-_{ij} \qquad \text{for all } i \text{ and } j,\ i \ne j$$
$$\beta_i - \beta_j = \beta^+_{ij} + \beta^-_{ij} \qquad \text{for all } i \text{ and } j,\ i \ne j$$
$$\alpha^+_{ij}, \alpha^-_{ij}, \beta^+_{ij}, \beta^-_{ij} \ge 0 \qquad \text{for all } i \text{ and } j,\ i \ne j$$

Coordinate conditions with respect to relative location assignment variables

$$x''_2 \le x'_1 + (1 - z^x_{12})M \qquad y''_2 \le y'_1 + (1 - z^y_{12})M$$
$$x''_1 \le x'_2 + (1 - z^x_{21})M \qquad y''_1 \le y'_2 + (1 - z^y_{21})M$$
$$x''_3 \le x'_1 + (1 - z^x_{13})M \qquad y''_3 \le y'_1 + (1 - z^y_{13})M$$
$$x''_1 \le x'_3 + (1 - z^x_{31})M \qquad y''_1 \le y'_3 + (1 - z^y_{31})M$$
$$x''_3 \le x'_2 + (1 - z^x_{23})M \qquad y''_3 \le y'_2 + (1 - z^y_{23})M$$
$$x''_2 \le x'_3 + (1 - z^x_{32})M \qquad y''_2 \le y'_3 + (1 - z^y_{32})M$$

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MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000) with building of 100 by 80.

$$x_j'' \le x_i' + (1 - z_{ij}^x)M \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$y_j'' \le y_i' + (1 - z_{ij}^y)M \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$z_{ij}^x + z_{ji}^x + z_{ij}^y + z_{ji}^y \ge 1 \quad \text{for all } i \text{ and } j,\ i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x_i', x_i'', y', y_i'' \ge 0 \quad \text{for all } i$$

$$z_{ij}^x, z_{ij}^y = \{0,1\} \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\alpha_i - \alpha_j = \alpha_{ij}^+ + \alpha_{ij}^- \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\beta_i - \beta_j = \beta_{ij}^+ + \beta_{ij}^- \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\alpha_{ij}^+, \alpha_{ij}^-, \beta_{ij}^+, \beta_{ij}^- \ge 0 \quad \text{for all } i \text{ and } j,\ i \ne j$$

No overlap between departments

$$z_{12}^x + z_{21}^x + z_{12}^y + z_{21}^y \ge 1$$

$$z_{13}^x + z_{31}^x + z_{13}^y + z_{31}^y \ge 1$$

$$z_{23}^x + z_{32}^x + z_{23}^y + z_{32}^y \ge 1$$

MIP-Model: Example

Assume 3 departments: A (3000) B (1000) C (4000)
with building of 100 by 80.

$$x''_j \le x'_i + (1 - z^x_{ij})M \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$y''_j \le y'_i + (1 - z^y_{ij})M \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$z^x_{ij} + z^x_{ji} + z^y_{ij} + z^y_{ji} \ge 1 \quad \text{for all } i \text{ and } j,\ i < j$$

$$\alpha_i, \beta_i \ge 0 \quad \text{for all } i$$

$$x'_i, x''_i, y', y''_i \ge 0 \quad \text{for all } i$$

$$z^x_{ij}, z^y_{ij} = \{0,1\} \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\alpha_i - \alpha_j = \alpha^+_{ij} - \alpha^-_{ij} \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\beta_i - \beta_j = \beta^+_{ij} - \beta^-_{ij} \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\alpha^+_{ij}, \alpha^-_{ij}, \beta^+_{ij}, \beta^-_{ij} \ge 0 \quad \text{for all } i \text{ and } j,\ i \ne j$$

$$\alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3 \ge 0$$

$$x'_1, x''_1, y'_1, y''_1, x'_2, x''_2, y'_2, y''_2, x'_2, x''_2, y'_2, y''_2 \ge 0$$

$$z^x_{12}, z^y_{12}, z^x_{21}, z^y_{21}, z^x_{13}, z^y_{13}, z^x_{31}, z^y_{31}, z^x_{23}, z^y_{23}, z^x_{32}, z^y_{32} = \{0,1\}$$

$$\alpha_1 - \alpha_2 = \alpha^+_{12} - \alpha^-_{12}$$

$$\alpha_1 - \alpha_3 = \alpha^+_{13} - \alpha^-_{13}$$

$$\alpha_2 - \alpha_3 = \alpha^+_{23} - \alpha^-_{23}$$

$$\beta_1 - \beta_2 = \beta^+_{12} - \beta^-_{12}$$

$$\beta_1 - \beta_3 = \beta^+_{13} - \beta^-_{13}$$

$$\beta_2 - \beta_3 = \beta^+_{23} - \beta^-_{23}$$

$$Minimize\ 10\left(|\alpha_1-\alpha_2|+|\beta_1-\beta_2|\right)$$
$$+20\left(|\alpha_1-\alpha_3|+|\beta_1-\beta_3|\right)+50\left(|\alpha_3-\alpha_2|+|\beta_3-\beta_2|\right)$$

$$37.5 \le x_1'' - x_1',\ x_1'' - x_1' \le 100$$
$$12.5 \le x_2'' - x_2',\ x_2'' - x_2' \le 40$$
$$50 \le x_3'' - x_3',\ x_3'' - x_3' \le 100$$

$$30 \le y_1'' - y_1',\ y_1'' - y_1' \le 80$$
$$10 \le y_2'' - y_2',\ y_2'' - y_2' \le 80$$
$$40 \le y_3'' - y_3',\ y_3'' - y_3' \le 80$$
$$(x_1''-x_1')(y_1''-y_1')=3000$$
$$(x_2''-x_2')(y_2''-y_2')=1000$$
$$(x_3''-x_3')(y_3''-y_3')=4000$$

$$x_1' \le x_1'' \quad y_1' \le y_1'' \quad \alpha_1 = 0.5x_1' + 0.5x_1''$$
$$x_2' \le x_2'' \quad y_2' \le y_2'' \quad \alpha_2 = 0.5x_2' + 0.5x_2''$$
$$x_3' \le x_3'' \quad y_3' \le y_3'' \quad \alpha_3 = 0.5x_3' + 0.5x_3''$$
$$x_1'' \le 100 \quad y_1'' \le 80 \quad \beta_1 = 0.5y_1' + 0.5y_1''$$
$$x_2'' \le 100 \quad y_1'' \le 80 \quad \beta_2 = 0.5y_2' + 0.5y_2''$$
$$x_3'' \le 100 \quad y_1'' \le 80 \quad \beta_3 = 0.5y_3' + 0.5y_3''$$

$$x_2'' \le x_1' + (1-z_{12}^x)M \quad y_2'' \le y_1' + (1-z_{12}^y)M$$
$$x_1'' \le x_2' + (1-z_{21}^x)M \quad y_1'' \le y_2' + (1-z_{21}^y)M$$
$$x_3'' \le x_1' + (1-z_{13}^x)M \quad y_3'' \le y_1' + (1-z_{13}^y)M$$
$$x_1'' \le x_3' + (1-z_{31}^x)M \quad y_1'' \le y_3' + (1-z_{31}^y)M$$
$$x_3'' \le x_1' + (1-z_{13}^x)M \quad y_3'' \le y_2' + (1-z_{23}^y)M$$
$$x_1'' \le x_3' + (1-z_{31}^x)M \quad y_2'' \le y_3' + (1-z_{32}^y)M$$

$$z_{12}^x + z_{21}^x + z_{12}^y + z_{21}^y + \ge 1$$
$$z_{13}^x + z_{31}^x + z_{13}^y + z_{31}^y + \ge 1$$
$$z_{23}^x + z_{32}^x + z_{23}^y + z_{32}^y + \ge 1$$

$$\alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3 \ge 0$$
$$x_1', x_1'', y'_1, y_1'', x_2', x_2'', y'_2, y_2'', x_2', x_2'', y'_2, y_2'' \ge 0$$
$$z_{12}^x, z_{12}^y, z_{21}^x, z_{21}^y, z_{13}^x, z_{13}^y, z_{31}^x, z_{31}^y, z_{23}^x, z_{23}^y, z_{32}^x, z_{32}^y = \{0,1\}$$

$$Minimize\ 10(\alpha_{12}^{+}+\alpha_{12}^{-}+\beta_{12}^{+}+\beta_{12}^{-})$$
$$+20(\alpha_{13}^{+}+\alpha_{13}^{-}+\beta_{13}^{+}+\beta_{13}^{-})+50(\alpha_{23}^{+}+\alpha_{23}^{-}+\beta_{23}^{+}+\beta_{23}^{-})$$

$$37.5 \le x_1'' - x_1', x_1'' - x_1' \le 100$$
$$12.5 \le x_2'' - x_2', x_2'' - x_2' \le 40$$
$$50 \le x_3'' - x_3', x_3'' - x_3' \le 100$$
$$30 \le y_1'' - y_1', y_1'' - y_1' \le 80$$
$$10 \le y_2'' - y_2', y_2'' - y_2' \le 80$$
$$40 \le y_3'' - y_3', y_3'' - y_3' \le 80$$
$$109 \le (x_1'' - x_1' + y_1'' - y_1') \le 130$$
$$63 \le (x_2'' - x_2' + y_2'' - y_2') \le 110$$
$$126 \le (x_3'' - x_3' + y_3'' - y_3') \le 140$$

$$x_1' \le x_1'' \quad y_1' \le y_1'' \quad \alpha_1 = 0.5x_1' + 0.5x_1''$$
$$x_2' \le x_2'' \quad y_2' \le y_2'' \quad \alpha_2 = 0.5x_2' + 0.5x_2''$$
$$x_3' \le x_3'' \quad y_3' \le y_3'' \quad \alpha_3 = 0.5x_3' + 0.5x_3''$$
$$x_1'' \le 100 \quad y_1'' \le 80 \quad \beta_1 = 0.5y_1' + 0.5y_1''$$
$$x_2'' \le 100 \quad y_1'' \le 80 \quad \beta_2 = 0.5y_2' + 0.5y_2''$$
$$x_3'' \le 100 \quad y_1'' \le 80 \quad \beta_3 = 0.5y_3' + 0.5y_3''$$

$$x_2'' \le x_1' + (1 - z_{12}^x)M \quad y_2'' \le y_1' + (1 - z_{12}^y)M$$
$$x_1'' \le x_2' + (1 - z_{21}^x)M \quad y_1'' \le y_2' + (1 - z_{21}^y)M$$
$$x_3'' \le x_1' + (1 - z_{13}^x)M \quad y_3'' \le y_1' + (1 - z_{13}^y)M$$
$$x_1'' \le x_3' + (1 - z_{31}^x)M \quad y_1'' \le y_3' + (1 - z_{31}^y)M$$
$$x_3'' \le x_1' + (1 - z_{13}^x)M \quad y_3'' \le y_2' + (1 - z_{23}^y)M$$
$$x_1'' \le x_3' + (1 - z_{31}^x)M \quad y_2'' \le y_3' + (1 - z_{32}^y)M$$

$$z_{12}^x + z_{21}^x + z_{12}^y + z_{21}^y + \ge 1$$
$$z_{13}^x + z_{31}^x + z_{13}^y + z_{31}^y + \ge 1$$
$$z_{23}^x + z_{32}^x + z_{23}^y + z_{32}^y + \ge 1$$

$$\alpha_1 - \alpha_2 = \alpha_{12}^{+} - \alpha_{12}^{-}$$
$$\alpha_1 - \alpha_3 = \alpha_{13}^{+} - \alpha_{13}^{-}$$
$$\alpha_2 - \alpha_3 = \alpha_{23}^{+} - \alpha_{23}^{-}$$
$$\beta_1 - \beta_2 = \beta_{12}^{+} - \beta_{12}^{-}$$
$$\beta_1 - \beta_3 = \beta_{13}^{+} - \beta_{13}^{-}$$
$$\beta_2 - \beta_3 = \beta_{23}^{+} - \beta_{23}^{-}$$

$$\alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3 \ge 0$$
$$x_1', x_1'', y'_1, y_1'', x_2', x_2'', y'_2, y_2'', x_2', x_2'', y'_2, y_2'' \ge 0$$
$$z_{12}^x, z_{12}^y, z_{21}^x, z_{21}^y, z_{13}^x, z_{13}^y, z_{31}^x, z_{31}^y, z_{23}^x, z_{23}^y, z_{32}^x, z_{32}^y = \{0,1\}$$

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How to Solve MIP?

See an overview here:

https://www.youtube.com/watch?v=i8WS6HIE8qM

Commercial Packages

Flow planner by Propplanner: https://www.proplanner.com/solutions/material-logistics-planning/flow-planner/

Layout-iQ™ by Rapid Modeling Corporation

Plant Design & Optimization by Siemens PLM Software: https://www.youtube.com/watch?v=ePZheUvsH0w; https://www.youtube.com/watch?v=8BTmNf3oCFY

Suite of software: FactoryCAD, FactoryFlow, FactoryMockup, and Plant Simulation

CORELAP

COmputerized RElationship LAyout Planning

Calculates a total closeness rating (TCR) from the relationship chart

$$\text{Layout score} = \sum_{AllDepartments} \text{numerical closeness rating}$$

Department with highest TCR is placed in the center

Tie-breaker is area size, then department number.

Chosen by the relationship with first dept.

Start with A relationships

If no A's, then E's, I's, etc.

CORELAP Example

A	= 64
E	= 16
I	= 4
O	= 1
U	= 0
X	= -1,024

			Relationship Chart							
Department Name	**Area (ft^2)**	**# of Grids**	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**
1. A:Receiving	12,000	30	-	I	U	O	U	U	U	U
2 B:Milling	8,000	20		-	U	I	O	O	U	U
3 C:Press	6,000	15			-	U	U	U	U	U
4 D:Screw m/c.	12,000	30				-	O	U	U	U
5 E:Assembly	8,000	20					-	A	O	U
6 F:Plating	12,000	30						-	E	U
7 G:Shipping	12,000	30							-	U
8 H:Dummy	2,000	5								-

CORELAP Example

A	= 64
E	= 16
I	= 4
O	= 1
U	= 0
X	= -1,024

Example:
https://youtu.be/hXvMoGLgeds?si=6QYhO_f0ZWxv7BlY

			Relationship Chart					
Department Name	**Area (ft^2)**	**# of Grids**	**A**	**E**	**I**	**O**	**U**	**TCR**
1. A:Receiving	12,000	30	0	0	1	1	5	**5**
2 B:Milling	8,000	20	0	0	2	2	3	**10**
3 C:Press	6,000	15	0	0	0	0	7	**0**
4 D:Screw m/c.	12,000	30	0	0	1	2	4	**6**
5 E:Assembly	8,000	20	1	0	0	3	3	**67**
6 F:Plating	12,000	30	1	1	0	1	4	**81**
7 G:Shipping	12,000	30	0	1	0	1	5	**17**
8 H:Dummy	2,000	5	0	0	0	0	7	**0**

Examples:

XYZ Inc. has a facility with six departments (A, B, C, D, E, and F). A summary of the processing sequence for 10 products and the weekly production forecasts for the products are given in the tables below.

a. Develop the from-to chart based on the expected weekly production.

b. Develop a block layout using SLP.

Product	Processing Sequence	Weekly Production
1	ABCDEF	960
2	ABCBEDCF	1200
3	ABCDEF	720
4	ABCEBCF	2400
5	ACEF	1800
6	ABCDEF	480
7	ABDECBF	2400
8	ABDECBF	3000
9	ABCDF	960
10	ABDEF	1200

Dept.	A	B	C	D	E	F
A	-					
B		-				
C			-			
D				-		
E					-	
F						-

XYZ Inc. has a facility with six departments (A, B, C, D, E, and F). A summary of the processing sequence for 10 products and the weekly production forecasts for the products are given in the tables below.

a. Develop the from-to chart based on the expected weekly production.

b. Develop a block layout using SLP.

Since we do not have the activity relationship chart, we base our closeness rating on the material flow information.

Dept.	A	B	C	D	E	F
A	-					
B		-				
C			-			
D				-		
E					-	
F						-

Dept.	A	B	C	D	E	F
A	-	A	U	U	U	U
B		-	A	E	E	I
C			-	O	E	O
D				-	I	U
E					-	I
F						-

Rank flow values:

1.	B-C	18,000	A
2.	A-B	13,320	A
3.	C-E	12,000	E
4.	B-E	10,320	E
5.	B-D	9,600	E
6.	D-E	6,240	I
7.	B-F	5,400	I
8.	E-F	5,160	I
9.	C-D	3,600	O
10.	C-F	3,600	O
11.	A-C	1,800	U
12.	D-F	960	U
13.	A-D	0	U
14.	A-E	0	U
15.	A-F	0	U

THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

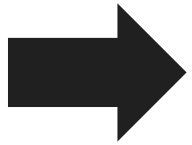
XYZ Inc. has a facility with six departments (A, B, C, D, E, and F). A summary of the processing sequence for 10 products and the weekly production forecasts for the products are given in the tables below.

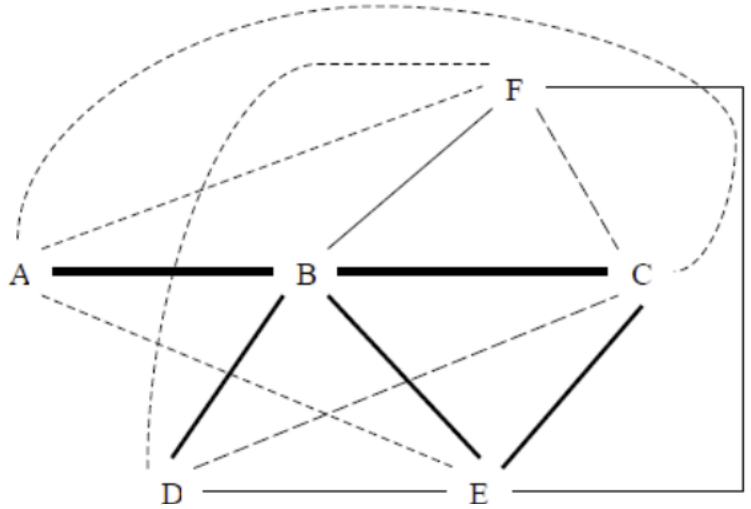
a. Develop the from-to chart based on the expected weekly production.

b. Develop a block layout using SLP.

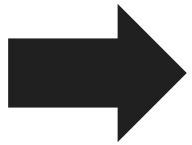
Since we do not have the activity relationship chart, we base our closeness rating on the material flow information.

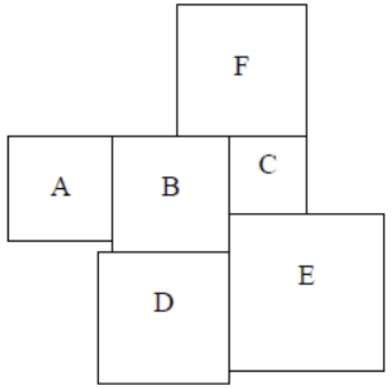
Dept.	A	B	C	D	E	F
A	-	A	U	U	U	U
B		-	A	E	E	I
C			-	O	E	O
D				-	I	U
E					-	I
F						-





Dept.	Dimension
A	40’ * 40’
B	45’ * 45’
C	30’ * 30’
D	50’ *50’
E	60’ * 60’
F	50’ *50’





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Suppose five departments labeled A through E are located as shown in the layout below. Given the corresponding flow-between chart, compute the efficiency rating for the layout.

A	B
C	
D	E

	A	B	C	D	E
A	—	5	0	4	−3
B		—	6	−1	2
C			—	−6	0
D				—	3
E					—

Suppose five departments labeled A through E are located as shown in the layout below. Given the corresponding flow-between chart, compute the efficiency rating for the layout.

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D				—	3
E					—

Positive adjacency **Negative adjacency**

$$Maximize\ Z = \frac{\sum_{(i,j)\in S=1}^{m} f_{ij}\, x_{ij} + \sum_{(i,j)\in \bar{S}=1}^{m} f_{ij}\,(1 - x_{ij})}{\sum_{(i,j)\in S=1}^{m} f_{ij} - \sum_{(i,j)\in \bar{S}=1}^{m} f_{ij}}$$

Positive department pairs **Negative department pairs**

Suppose five departments labeled A through E are located as shown in the layout below. Given the corresponding flow-between chart, compute the efficiency rating for the layout.

A	B
C	
D	E

	A	B	C	D	E
A	—	5	0	4	−3
B		—	6	−1	2
C			—	−6	0
D				—	3
E					—

$$Maximize\ Z = \frac{\sum_{(i,j)\in S=1}^{m} f_{ij}\, x_{ij} + \sum_{(i,j)\in \bar{S}=1}^{m} f_{ij}\,(1 - x_{ij})}{\sum_{(i,j)\in S=1}^{m} f_{ij} - \sum_{(i,j)\in \bar{S}=1}^{m} f_{ij}}$$

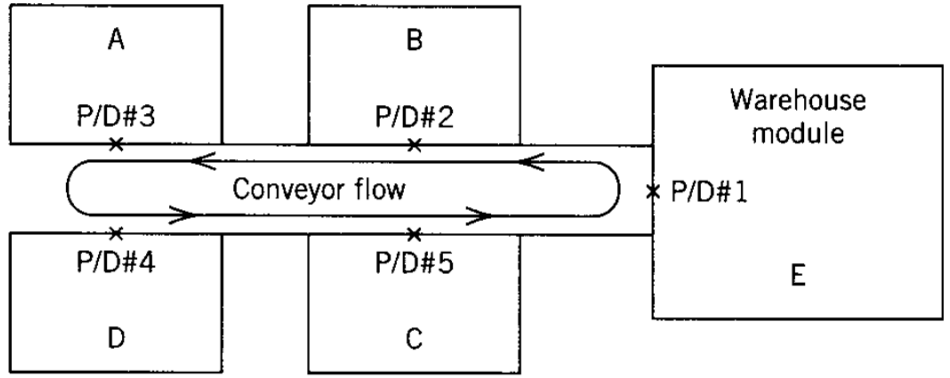
$$\sum_{(i,j)\in F} f_{ij}\, x_{ij} = 14, \qquad \sum_{(i,j)\in \bar{F}} f_{ij}\,(1 - x_{ij}) = -4;$$

$$\sum_{(i,j)\in F} f_{ij} = 20, \qquad \sum_{(i,j)\in \bar{F}} f_{ij} = -10,$$

$$z = (14 - (-4))/(20 - (-10)) = 18/30 = 0.60.$$

In an assembly plant, material handling between departments is performed using a unidirectional closed-loop conveyor. The figure below shows the layout for the modular facility, which consists of three equal-sized assembly modules (A, B, and C), one administrative module (D), and one warehouse module (E). P/D points for each module are also shown in the figure. The administrative and warehouse activities are not to be moved; however, assembly areas A, B, C can be relocated. The distance between P/D points and the number of pallet loads moved between departments are given below.

Using the pairwise exchange method, determine new locations for assembly modules A, B, and C that minimize the sum of the products of pallet flows and conveyor travel distances.



Distance between P/Ds

From	To	Distance
P/D 1	P/D 2	60′
P/D 2	P/D 3	90′
P/D 3	P/D 4	30′
P/D 4	P/D 5	90′
P/D 5	P/D 1	60′

Pallet Flow per Day

From/To	A	B	C	D	E
A	0	0	5	0	30
B	10	0	25	0	0
C	25	5	0	0	0
D	0	0	0	0	0
E	5	20	5	0	0

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Using the pairwise exchange method, determine new locations for assembly modules A, B, and C that minimize the sum of the products of pallet flows and conveyor travel distances.

Current costs: A to PD #3, B to PD #2, C to PD #5

$$\text{Cost} = 5(120) + 30(180) + 10(90) + 25(210) + 25(210) + 5(120) + 5(150) + 20(60) + 5(270) = 21{,}300$$

Exchange 2-3-5:

$$\text{Cost} = 5(210) + 30(270) + 10(240) + 25(120) + 25(120) + 5(210) + 5(60) + 20(150) + 5(60) = 22{,}200$$

Exchange 5-2-3:

$$\text{Cost} = 5(210) + 30(60) + 10(210) + 25(90) + 25(120) + 5(240) + 5(270) + 20(60) + 5(180) = 14{,}850$$

Exchange 3-5-2:

$$\text{Cost} = 5(240) + 30(180) + 10(210) + 25(120) + 25(90) + 5(210) + 5(150) + 20(270) + 5(60) = 21{,}450$$

Exchange 2-5-3:

$$\text{Cost} = 5(90) + 30(270) + 10(120) + 25(210) + 25(240) + 5(120) + 5(60) + 20(270) + 5(210) = 28{,}350$$

Exchange 5-3-2:

$$\text{Cost} = 5(120) + 30(60) + 10(120) + 25(240) + 25(210) + 5(90) + 5(270) + 20(150) + 5(60) = 19{,}950$$

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